

An Initial Assessment of Variable Depth Liner Optimization for Ducted Proprotor Applications

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The potential rise of the Urban Air Mobility market has spurred the design of a new generation of novel aircraft. To aid industry and researchers interested in these types of aircraft, the Revolutionary Vertical Lift Technology project at NASA has developed a fleet of reference vehicles for system studies. A new six-passenger reference vehicle has recently been added to the research portfolio that has ducted proprotors for propulsors. The ducts present an opportunity to apply acoustic treatment to the interior of the duct that could target both tonal and broadband noise, representative of the sound produced by this type of propulsor. In this paper, methodologies are presented to design a variable depth liner for this application. An optimizer is used to design multiple liners with variable chamber depths. Experimental results from normal impedance testing are compared to numerical predictions using the optimizer model and a finite element model. Results show that an optimizer can be used to design an acoustic liner with favorable performance for a broad range of frequencies, which could be appropriate for a ducted proprotor application.

I. Introduction

Urban Air Mobility (UAM) has emerged as a rising topic in the aviation industry and has garnered interest from airframers and researchers alike. However, the success of UAM aircraft and missions is largely dependent on public acceptability. One important factor to public acceptability of the class of aircraft proposed for the UAM ecosystem is aircraft noise. As these vehicles will likely produce sound that is both tonal and broadband in nature, it is vital to implement devices capable of noise reduction for both types of sound.

To provide a platform for research to advance the UAM mission, researchers developed reference vehicles for the Revolutionary Vertical Lift Technology (RVLT) project's research portfolio [1–4] including a Tiltduct reference vehicle [5]. This concept aircraft is powered by six proprotors encased in individual ducts that tilt to transition from hover flight to cruise flight.

A ducted proprotor generates noise that is likely both tonal and broadband in nature. The proprotors in particular would generate strong tonal noise at the blade passage frequency (BPF) and associated harmonics. For the Tiltduct reference vehicle, the BPF could be about 200 Hz or lower for an eight-bladed proprotor with a diameter of ~7 ft and a blade tip speed of 550 ft/s [5]. Broadband noise would be generated by the proprotors and inlet flow distortions. The noise generated by the proprotors and ducts would likely increase during the transition from hover to forward flight due to nonuniform flow. While there are aerodynamic and weight challenges for this design, there could be an opportunity to apply acoustic treatment to these ducts to target both tonal and broadband noise. Sound attenuation devices known as acoustic liners are currently used in conventional air transport engine nacelles and could be used to acoustically treat these proprotor ducts to reduce generated noise. The concept proposed for this paper is a variable depth liner, in which the chambers of the acoustic liner vary in length to attenuate sound at multiple frequencies. There has been much research in this topic as many researchers have developed their own types of variable depth liners.

In the 1970s, Wirt [6] presented a concept called Parasone that is comprised of “waveguidelets” of various depths. While he developed many different sound absorbing concepts, Parasone is the concept that is most like the variable depth liners analyzed in this work. Most of Wirt's concepts varied the chamber length to modify the frequency range of the reactive absorbers. In 2012, Jones et al. [7] presented an evaluation of variable depth liners for a no flow condition. In 2013, Hua et al. [8] combined a microperforated panel (i.e., a perforated facesheet) with a substrate (i.e., a liner core) based on Wirt's Parasone concept to design an acoustic liner with improved absorption at low frequencies while

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still providing broadband absorption. In 2015, Jones et al. [9] extended their work on variable depth liners to a flow condition in which they compared experimental results from the NASA Langley Grazing Flow Impedance Tube [10] to numerical results. They also employed an optimization procedure to design a new liner to fit the geometry of the NASA Langley Curved Duct Test Rig. In 2018, Schiller and Jones [11] evaluated a smeared impedance model for variable depth liners. They demonstrated that semianalytical models could, with some corrections, be used to predict impedance spectra and additionally, be used as a model in an optimizer for liner design. In 2021, Palani et al. [12, 13] presented work on variable depth chamber liners (that had folded chambers) to provide broadband sound absorption.

While only a cursory review of some literature is provided, it shows that there is considerable interest in designing acoustic liners with variable depth chambers for increased performance. Reducing noise in the turbofan engine nacelles in conventional aircraft has been the primary motivation for variable depth liner research. The motivation for the current work is the design of acoustic liners for use in a propotor duct. For such an application, it seems clear that the frequencies of interest will be lower and the available volume will be reduced relative to a conventional turbofan application. Variable depth liners that include bent chambers offer the ability to achieve both low frequency and broadband attenuation.

The NASA Liner Physics Team has developed a new liner optimization methodology that uses the semianalytical impedance model from Schiller and Jones [11]. Thus, the goal of this investigation is to validate this optimization approach for variable depth liners using experimental data acquired in the NASA Langley Normal Incidence Tube (NIT). The target frequency range for this study is set to be consistent with those used in the NIT. In addition, straight chamber configurations (no bent chambers) are considered to minimize liner complexity in this initial assessment.

The rest of this paper is organized as follows. First, Section II describes the numerical setup for this work, which includes the methodology and models used for designing variable depth liners. Section III describes the experimental setup, including the testing methods and a description of the test articles developed for this work. Then, Section IV provides results from the predictions and experiments, followed by concluding remarks in Section V.

II. Numerical Setup

In this section, the optimization methodology and impedance prediction models used to design a variable depth liner, as shown in Figure 1, are presented. Two impedance prediction models are described. The first is a semianalytical impedance model that is used in the optimization to design the acoustic liners. The second is a finite element model that provides higher fidelity predictions for comparison against the semianalytical impedance model predictions and experimental data.

A. Optimization Methodology

A differential evolution optimizer [14] (implemented in SciPy) was used to design liner samples for this work. Five optimizing arguments were provided to the optimizer, which corresponded to five chamber depths of the acoustic liner. The lower and maximum bounds of chamber depths imposed on the optimizer were 0.01" and 4.00", respectively. The maximum bound was chosen based on test sample size limitations for future testing in the NASA Langley Grazing Flow Impedance Tube. Initial guesses for the chamber lengths were calculated using quarter-wavelength theory to target the following five frequencies: 900, 1250, 1600, 1950, and 2500 Hz. Quarter-wavelength theory states that the chamber length should be

$$L = \frac{1}{4}\lambda = \frac{1}{4}\frac{c}{f}, \quad (1)$$

in which L is the resonator chamber length, λ is the acoustic wavelength, $c = 343$ m/s is the sound speed, and f is the targeted frequency. The initial guesses for the chamber lengths were calculated as 3.751", 2.701", 2.110", 1.731", and 1.350". To determine if there is a better set of chamber lengths, this optimizer seeks to minimize the objective function, F , as described by Nark and Jones [15]:

$$F = \left\{ \sum_{i=1}^{N_f} W_i [\zeta_{opt}(f_i) - \zeta_{pred}(f_i)] [\zeta_{opt}(f_i) - \zeta_{pred}(f_i)]^* \right\}^{0.5}, \quad (2)$$

in which a weighting factor W is applied to N_f frequencies f . The optimal impedance is $\zeta_{opt}(f)$ and the predicted impedance of the liner from the semianalytical impedance model is $\zeta_{pred}(f)$. Note that the asterisk denotes the complex conjugate. The optimal impedance is $1.0 + i0.0$ at all frequencies for plane waves in the Normal Incidence

Tube but when designing samples without a facesheet, an optimal impedance of $0.1 + i0.0$ was chosen, which will be discussed in further detail later. For this initial work, all frequencies were equally weighted ($W = 1$).

There are additional key parameters used in differential evolution optimizers. The iteration limit was set to 1000, however, this limit was never reached during this study. The population size multiplier, mutation constant, and recombination constant were set to 50, 0.5, and 0.7, respectively. The relative tolerance for convergence was set to 0.01.

B. Semianalytical Impedance Model

The semianalytical impedance model (SIM) is comprised of two parts: the Zwikker-Kosten Transmission Line (ZKTL) model and a facesheet model (FSM). The ZKTL model was developed based on analysis conducted by Zwikker and Kosten [16] and uses a transmission line formulation to calculate the acoustic pressure, acoustic particle velocity, and normalized impedance* at the end of a resonator chamber [10, 17, 18]. The FSM is a lumped parameter model used to predict the change in impedance across a perforated facesheet [19, 20].

Figure 1 shows a variable depth liner with five different chambers and a perforated facesheet (shown in blue) that has a thickness t . The boundary between the chamber and facesheet is represented by a thick green dotted line and the surface of the perforated facesheet is represented by a thick red dashed line. The active area of the chamber is bounded by dashed orange lines. This representation has two holes per chamber.

At the end of each chamber there is a hard wall backplate in which the acoustic pressure is $p_0 = 1$ and acoustic particle velocity is $v_0 = 0$, normalized by ρc^2 and c , respectively. The ZKTL model is used to calculate the pressure and velocity (p_1 and v_1 in Fig. 1) at the facesheet boundary, from which the impedance ζ_1 is calculated. This impedance is used as an input to the FSM to predict the surface impedance $\zeta_{fs,1}$, above the facesheet. The N_c number of surface admittances (i.e., the inverse of the impedance) are summed to calculate the smeared impedance across the liner surface, ζ_s , as

$$\zeta_s = \left[\sum_{i=1}^{N_c} \Omega_i \beta_{fs,i} \right]^{-1}, \quad (3)$$

in which Ω_i and $\beta_{fs,i}$ are the surface open area ratio and surface admittance of the i th chamber, respectively. The smeared surface impedance of the entire liner is comprised of the resistance θ and reactance χ ,

$$\zeta_s = \theta + i\chi. \quad (4)$$

The SIM has a very low computational cost (less than a second) due to its semianalytical nature, which makes it a good model for parameter studies and optimizers. As such, this model is used in the optimizer for this work, as mentioned previously.

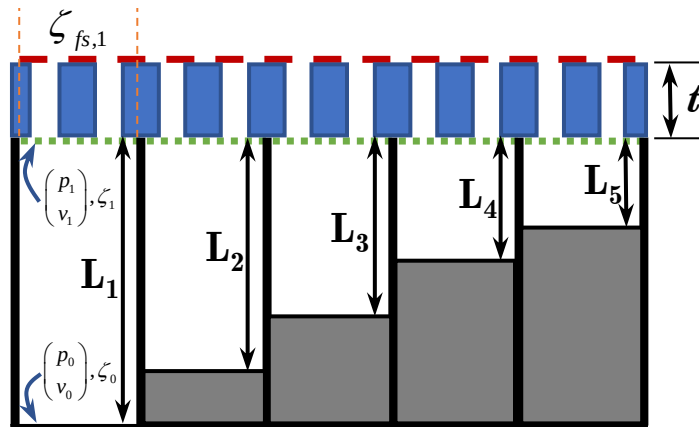


Fig. 1 A variable depth liner with five different chamber depths with a facesheet of thickness, t , applied.

*All impedances, resistances, reactances, and admittances discussed in this paper are normalized, i.e., nondimensionalized by ρc or $1/\rho c$ where appropriate.

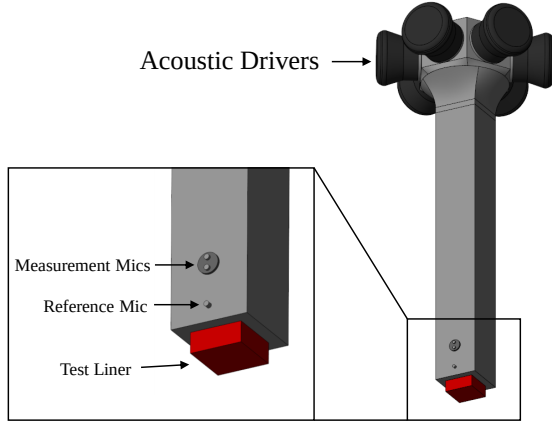


Fig. 2 Schematic of the NIT testing apparatus.

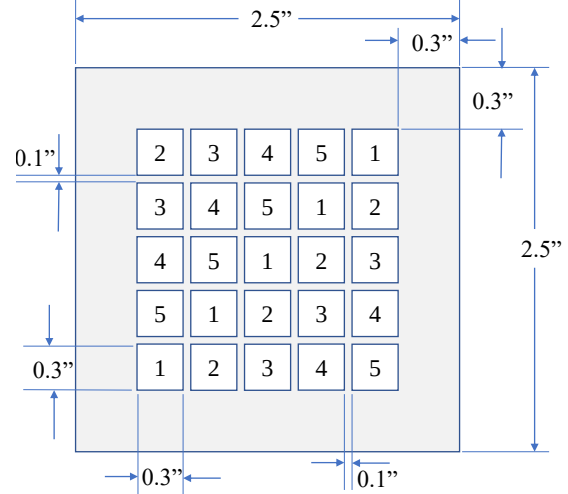


Fig. 3 Top view of liner cores presented in this work.

C. Finite Element Model

The second model used in this study is a finite element model (FEM) created by Schiller and Jones [11] using COMSOL Multiphysics® [21]. A benefit of the FEM is that higher fidelity predictions can be compared to the SIM and experimental data. Additionally, interactions between the chambers can be observed with the FEM. A significant disadvantage of the FEM is the computation time, which is at least two orders of magnitude longer than that of the SIM.

To model the facesheet in the FEM, an interior impedance boundary condition is applied at the end of each chamber (as represented by the dotted green lines in Fig. 1) that uses a modified version of the FSM. As noted in Refs. [11] and [22], the FSM was developed for perforated facesheets in the presence of grazing flow. The samples in this work were tested without grazing flow, and thus, some inaccuracies are expected. To account for this, engineering judgment was used to increase the discharge coefficient from 0.771 to 1.5×0.771 and the semiempirical constant from 64 to 5×64 to smooth out the reactances [22, p. 15]. Note that the modifications in Schiller et al. [22] were made specific to their configuration and certainly require additional investigation for the configurations presented in this work. Otherwise, the FEM and FSM are implemented as described by Schiller and Jones [11] and Jones et al. [18], respectively.

III. Experimental Setup

The experimental setup and test articles designed to validate the optimization approach will be presented in this section. For this initial assessment, samples were designed for testing in the Normal Incidence Tube at the NASA Langley Research Center.

A. Test Apparatus

The Normal Incidence Tube (NIT), shown in Figure 2, is a vertically oriented test apparatus and consists of four main components: a set of six loudspeakers[†] mounted to a manifold that combines six pipes to one, a waveguide, a set of microphones installed in the waveguide, and an adapter plate for mounting test liners. Note that the waveguide has a square profile with interior dimensions of $2" \times 2"$ and the adapter plate accepts square test articles with cross-sectional dimensions of $2.5" \times 2.5"$. The active area of the test article is $2" \times 2"$ to match the size of the waveguide. Two of the waveguide microphones[‡] are installed into a plug that can rotate via a stepper motor to precisely swap the positions of the microphones. These microphones are labeled “Measurement Mics” in Figure 2. The third microphone is a reference microphone (labeled “Reference Mic” in Fig. 2) that is used to measure the sound pressure level (SPL) near the surface of the test liner for the control system. Acoustic impedances were deduced using the method described by Chung and Blaser [23] using a swept sine source with a source SPL of 120 dB (re. $20 \mu\text{Pa}$). The reader is referred to previous literature [10, 24–28] for further details regarding the NIT.

[†]All loudspeakers are JBL 2446H high frequency compression drivers.

[‡]All microphones are Brüel & Kjær Type 4938 $\frac{1}{4}"$ pressure-field microphones.

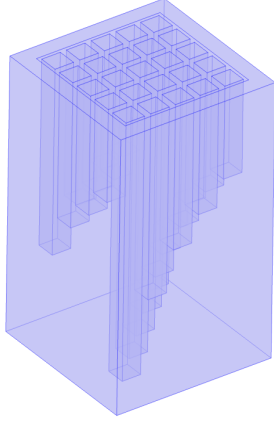


Fig. 4 CAD Model of a liner core.

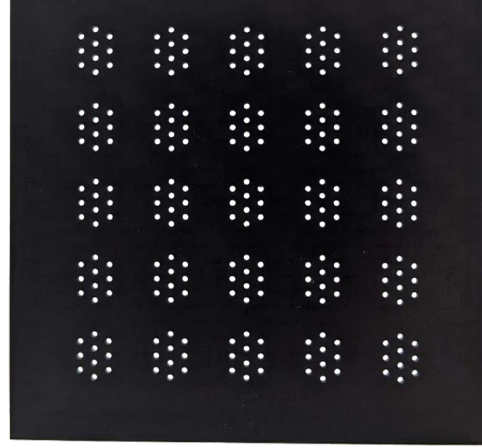


Fig. 5 The facesheet used in Configurations 3 and 4. Each group of 10 perforations covers a single chamber of the liner core.

B. Test Configurations

Four liner configurations were considered in this study, two without facesheets and two with. Each liner was designed to target the following five frequencies: 900, 1250, 1600, 1950, and 2500 Hz. A row of chambers contains the five chambers designed to target select frequencies as shown in Figure 3. These rows are repeated to form 25 total chambers; note that the chamber order is staggered in consecutive rows so that no two identical chambers are located next to each other, as shown in Figures 3 and 4.

The liner cores share all geometric dimensions, aside from the chamber lengths and overall length. Each chamber has cross-sectional dimensions of $0.3" \times 0.3"$, as shown in Figure 3. All chambers have backing that is at least 0.25" thick. The partitions between chambers are 0.1" wide and the perimeter of the outside chambers has a 0.3" border. The active area of the test article ($2.0" \times 2.0"$) is exposed to the waveguide and is ensounded during the normal incidence sound test. Each test article was 3D printed with Accura[®] 60 plastic using a stereolithography (SLA) machine.

Configuration 1 is a liner core designed using quarter-wavelength theory (see Eq. 1) with chamber lengths of 3.751", 2.701", 2.110", 1.731", and 1.350".

Configuration 2 is a liner core for which the chamber lengths were designed using the optimization approach described in Section II. While the ideal normal incidence impedance for sound attenuation of plane waves is $1.0 + i0.0$, this impedance cannot be obtained for this type of liner without the aid of a facesheet to add resistance. Preliminary testing of Configuration 1 showed that the resistance was about 0.1 over a large portion of the frequency range. Thus, engineering judgment was used to reduce the target impedance for this design to $0.1 + i0.0$ at each of the five target frequencies. With this modified goal, the optimizer selected the following five chamber depths: 3.716", 2.680", 2.096", 1.719", and 1.342".

Configuration 3 contains the same liner core as Configuration 1 but includes a perforated facesheet. Figure 5 shows a photograph of the facesheet, which has a thickness of 0.032", a hole diameter of 0.029", 10 holes per liner chamber, and a chamber open area ratio of 0.073.

Configuration 4 consists of a liner core in combination with the same perforated facesheet, in which the liner core was designed using the optimizer. The optimizer targeted a desired impedance of $1.0 + i0.0$ at each of the five frequencies. Five chamber depths were selected: 3.285", 3.283", 1.801", 1.243", and 1.174". Note that two of the chamber depths are almost identical, as the optimizer found that only four distinct chamber depths were needed when combined with this facesheet.

IV. Results

In this section, impedance and absorption coefficient spectra predicted using the SIM and the FEM, and deduced using experimental data for a uniform depth liner and the four liner configurations described in Section III will be presented.

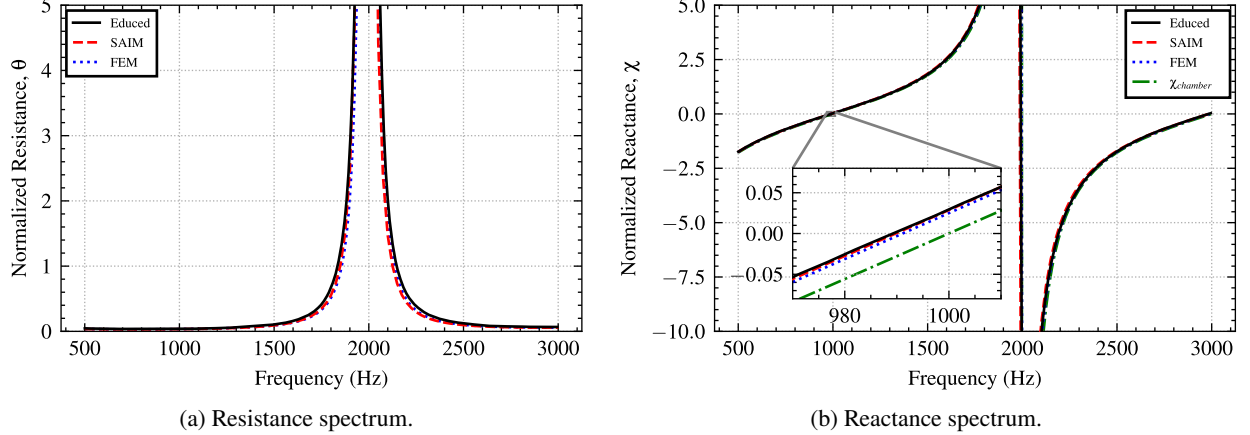


Fig. 6 (a) Resistance and (b) reactance spectra are shown for a uniform depth liner designed using quarter wavelength theory to target 1000 Hz.

A. Uniform Liner without a Facesheet

Figure 6 shows the (a) resistance and (b) reactance spectra for a uniform depth liner that was designed to have a resonant frequency of 1000 Hz (chambers are all 3.376" long). Observe that both SIM and FEM predictions compare favorably with the measured resistance and reactance spectra. The predicted and measured reactances compare favorably at the resonant frequency (1000 Hz), where the reactance crosses the zero axis with a positive slope. Additionally, both models predict the antiresonance at 2000 Hz, where an asymptote can be observed.

An additional reactance curve is included in Figure 6(b) (shown in dashed-dotted green) based on simple quarter-wavelength theory, i.e.,

$$\chi_{chamber} = - \left[\sum_{i=1}^{N_c} \sigma_i \tan(kL_i) \right]^{-1}, \quad (5)$$

in which $\chi_{chamber}$ is the chamber reactance, σ is the open area ratio of a single chamber, N_c is the number of chambers considered, and L is the chamber length. For this uniform depth liner, the open area ratio for one chamber is 0.02. This model ignores the effects of frictional losses along the walls of the chamber. Nevertheless, the model predicts reactances that compare favorably with the measured results at resonance and at antiresonance. This suggests that frictional losses along the wall of this uniform depth liner are minimal. Note that all data in Figure 6(b) show a second resonant frequency at 3000 Hz at the edge the reactance plot. These results are as expected, but this point will be important to note when analyzing the results of the variable depth configurations.

B. Variable Depth Liners without Facesheets

1. Configuration 1

Figure 7[§] shows the (a) resistance, (b) reactance, and (c) absorption coefficient spectra for Configuration 1, designed using quarter-wavelength theory targeting 900, 1250, 1600, 1950, and 2500 Hz. Note that the reactance spectrum contains a combination of $-\cot(kL)$ shapes for each chamber. When antiresonant behavior in the chamber is pronounced, there is a corresponding sharp increase in resistance. Intuitively, this makes sense as resistance is inversely proportional to acoustic particle velocity and the acoustic particle velocity is zero at antiresonances.

While quarter-wavelength theory accurately predicts the measured response for the uniform liner with no facesheet, the same is not true when the chamber depths are allowed to vary. As expected, there are five resonances (five zero crossings with positive slope) in the measured reactance spectrum. These resonances occur at frequencies just below those corresponding to quarter wavelengths: 873, 1206, 1527, 1846, and 2339 Hz. Note that the deviation from the target resonant frequency increases as the frequency increases.

[§]For most figures in this paper (Figs. 6, 7, 8, 10, and 11) that present impedance or absorption spectra, the plots will show educed results in solid black, the SIM results in dashed red, the FEM results in dotted blue, and quarter-wavelength calculations in dashed-dotted green.

The first possible explanation for this downward frequency shift is that the chambers could be deeper than intended. However, quality assurance checks of the liner core revealed that the actual chamber depths were on average 0.003" larger than the designed depths. This difference is insufficient to explain the deviation in resonant frequencies observed. The measured frequency shift could also be expected if the backplate is compliant. However, the impedance for the uniform depth liner was well predicted by both models. If the material used to build these two liners was not sufficiently rigid, the models (which assume a rigid backplate) would have been unable to properly predict the results for the uniform liner. Hence, it seems unlikely that the backplate is sufficiently compliant to explain the measured frequency shift.

Observe that the FEM predicts the measured impedance spectra quite well, except for increased magnitudes at the resonant and antiresonant frequencies. Therefore, the authors surmise that the frequency shift is caused by acoustic interactions between the chambers that are captured with the higher fidelity FEM. These interaction effects are not captured by the simpler quarter-wavelength model or the SIM. Further investigation of this effect will be the focus of future work.

The absorption coefficient α spectrum shown in Figure 7(c) is calculated as [19, p. 172]

$$\alpha(f) = \frac{4\theta(f)}{(\theta(f) + 1)^2 + \chi(f)^2} . \quad (6)$$

Note that this liner absorbs at least 40% ($\alpha > 0.4$) of the incoming sound for most frequencies between 850 and 2700 Hz and has a particularly large absorption at approximately 2600 Hz. Since this configuration has no facesheet to provide resistance, the resistance is well below unity for much of the frequency range presented. However, the reactance shows many frequencies (five resonances) where the reactance is ideal, i.e., zero. These very low resistances combine with the near-ideal reactances to enable this amount of sound absorption.

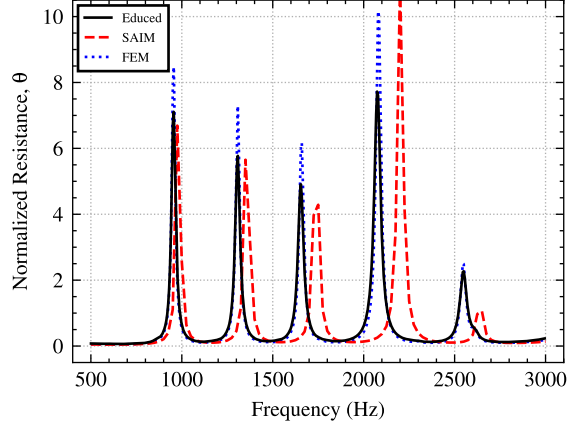
The absorption coefficient spectrum predicted with the FEM compares favorably with the measured data below 1500 Hz and above 2500 Hz. However, the FEM predictions deviate from the measured data between 1500 and 2500 Hz. This deviation is due to the slight differences in the predicted and measured impedances that are difficult to discern at the scale used in these figures.

The SIM predicted impedance demonstrates the same features found in the measured results. However, the predicted resonant and antiresonant frequencies are shifted up in frequency relative to the measured data. The shift becomes more pronounced with increasing frequency and can be loosely described as a “stretching” of the curve. These frequency shifts in the predicted resistance and reactance spectra have limited effect on the predicted absorption coefficient spectrum. This is at least partially due to the characteristics of the variable depth concept, whereby the absorption between resonances is increased if the two resonant frequencies are sufficiently close together. Therefore, although the SIM has limitations (e.g., failure to capture acoustic interactions between the chambers), it compares favorably with measured absorption. These results, coupled with its computational efficiency, demonstrate the suitability of the SIM for this liner design process.

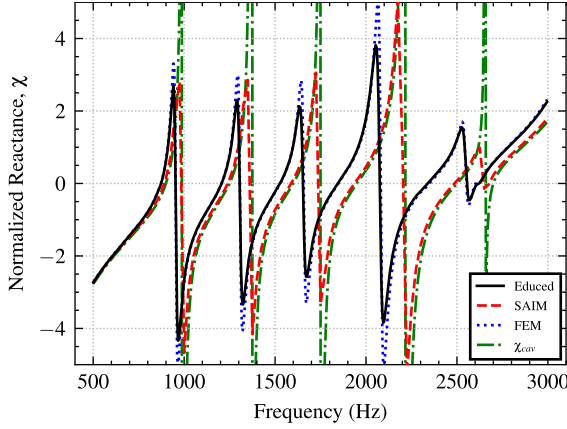
2. Configuration 2

Figure 8 shows the (a) resistance, (b) reactance, and (c) absorption coefficient spectra for Configuration 2, designed using the optimization approach. The measured resonant frequencies for Configuration 2 are 882, 1208, 1534, 1852, and 2336 Hz. These frequencies differ from those observed for Configuration 1 but remain lower than those predicted using quarter-wavelength theory. Two clear differences are present in the two configurations – the particular chamber lengths and the chamber layout. In Configuration 1, the chambers were arranged in rows of descending chamber lengths, offset so that no two identical chamber lengths were next to one another. In Configuration 2, each row contained one of each chamber length, but the layout was randomized as determined by the optimizer. The resistance for Configuration 2 is almost the same as that observed for Configuration 1 with the exception that the magnitude of the fifth antiresonance is ~0.5 larger in resistance and 10 Hz higher in frequency. A small bump in resistance is observed between the first and second antiresonance at ~1175 Hz and between the second and third antiresonance at ~1450 Hz. The measured reactance is also very similar to that of Configuration 1, with the exception that the fifth antiresonance has larger magnitude peaks and is located at a slightly higher frequency relative to the fifth antiresonance of Configuration 1. Regardless, the absorption coefficient spectrum matches the corresponding spectrum of Configuration 1 quite well, aside from small bumps at ~1175 and ~1450 Hz that result from the small bumps in resistance mentioned earlier.

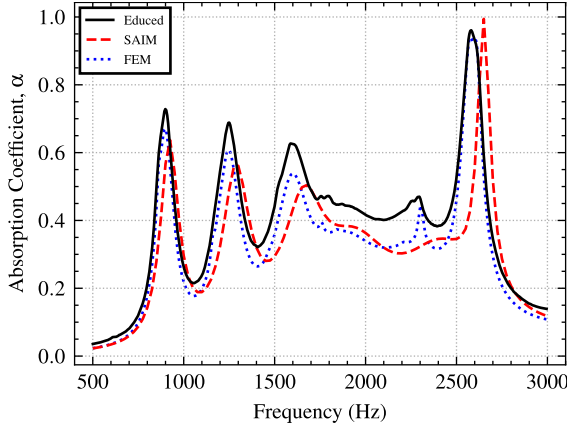
The FEM predictions match the measured data reasonably well, as was the case with Configuration 1, but they again overpredict the magnitude of the resistance and reactance at the antiresonances. Note that the FEM calculates the small increase in resistance at ~1175 and ~1450 Hz, which are hypothesized to be due to acoustic interactions between the



(a) Resistance spectrum.

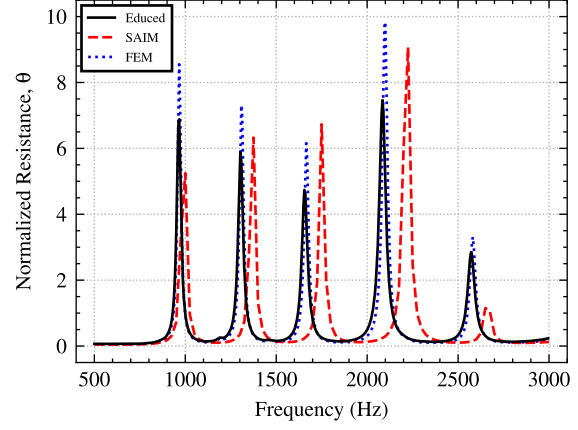


(b) Reactance spectrum.

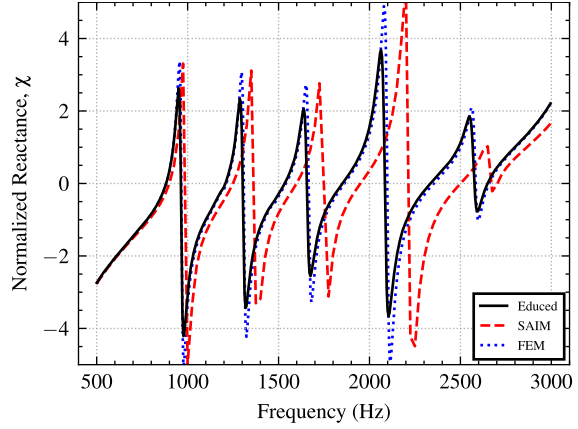


(c) Absorption coefficient spectrum.

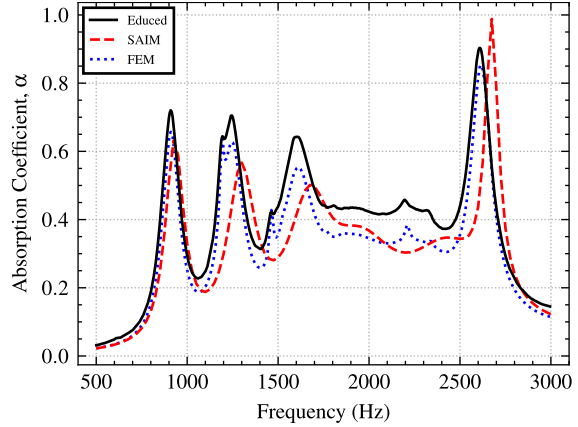
Fig. 7 Configuration 1 spectra are shown for (a) resistance, (b) reactance, and (c) absorption coefficient.



(a) Resistance spectrum.



(b) Reactance spectrum.



(c) Absorption coefficient spectrum.

Fig. 8 Configuration 2 spectra are shown for (a) resistance, (b) reactance, and (c) absorption coefficient.

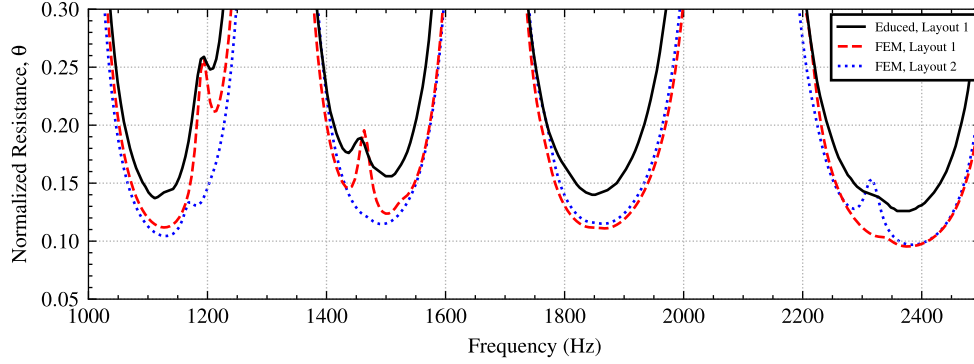


Fig. 9 Resistance spectrum from the educed and FEM results for the Configuration 2 layout compared to FEM results for an alternative arrangement of chambers.

chambers. To investigate these interactions, a second FEM was created with the same chamber depths as Configuration 2 but with a different arrangement of the chambers. Figure 9 shows an expanded view of the resistance spectrum educed from the experimental data and FEM results for Configuration 2 (from Fig. 8(a)) compared to the FEM resistance predictions for the alternative chamber arrangement. For this revised layout, the FEM simulation (dotted blue in Fig. 9) predicts a small increase in resistance at ~ 1160 Hz and a slightly larger increase in resistance at ~ 2310 Hz. The effects of chamber layout could be included to obtain the best results, but it appears these effects are not significant for the liners used in this study.

C. Variable Depth Liners with Perforated Facesheets

1. Configuration 3

This configuration uses the same liner core as Configuration 1 but includes a perforated facesheet as described in Section III. Since the inclusion of this facesheet makes this configuration a nonlinear liner, the results presented herein were acquired at a source SPL of 120 dB where nonlinear effects should not be significant.

Figure 10 shows the (a) resistance, (b) reactance, and (c) absorption coefficient spectra for Configuration 3. Adding the facesheet causes the resistance of the liner to increase considerably compared to the resistance of Configuration 1. Between 700 and 1700 Hz, the resistance is about 1.0, as compared to approximately 0.1 for Configuration 1.

The first three antiresonances observed in the reactance of Configuration 1 have been smoothed out by the perforated facesheet. Additionally, the facesheet has caused the reactance to increase with frequency, creating a positive slope in the reactance spectrum in Figure 10(b). For the frequencies presented, there is only one resonance at 1332 Hz. The positive slope is partially due to the mass reactance provided by the perforated facesheet, which increases with frequency [19, p. 181].

This liner provides excellent broadband absorption ($\alpha \geq 0.8$) from approximately 725 to 1750 Hz, and for a good portion of this frequency range the absorption coefficient is above 0.9. However, this liner provides less absorption from 1750 to 2250 Hz, at which it drops to around 0.4. A peak in absorption occurs at approximately 2300 Hz, due to the sharp decrease in reactance at this frequency. The absorption seems to peak again toward 3000 Hz, near an antiresonance. Overall, the addition of the perforated facesheet makes this configuration a better sound absorber than Configuration 1 at most frequencies considered.

The FEM (with a modified version of the FSM for the perforated facesheet) predicts the measured resistance and the reactance of this configuration well. However, it seems that the entire resistance spectrum predicted by the FEM is shifted by 50 Hz when compared to the experimental data shown in Figure 10(a). This shift is observed in the reactance spectrum as well, however, it seems that the shift in reactance is approximately 100 Hz. Additionally, the reactance spectrum of the FEM predictions seems to be much smoother from approximately 775 to 1500 Hz. The FEM predicts the absorption coefficient reasonably well, aside from the shifts in frequency and an overprediction of the absorption at approximately 875 Hz. This overprediction is due to the smooth resistance curve that the FEM predicts in this frequency range.

The SIM model (with an unmodified version of the FSM for the perforated facesheet) captures the trend of the resistance and reactance well. However, there are some notable discrepancies in the resistance. First, the entire resistance

spectrum predicted by the SIM seems to be shifted 25 Hz to the left. Also, the model overpredicts the two peaks in the measured resistance and the peaks seem to be narrower than the peaks observed experimentally. The frequency shift is seen in the reactance spectrum as well, as are overpredictions of the magnitude of the reactance peaks. In spite of the frequency shifts observed, the SIM seems to provide a better prediction of the absorption coefficient than the FEM.

2. Configuration 4

The liner core for this configuration was designed using the optimizer described in Section II with a target impedance of $1.0 + i0.0$. Note that the geometric properties of the facesheet were fixed and not selected by the optimizer. Unlike Configuration 2, in which all five chambers have distinctly different depths, the optimization results in two chambers with nearly the same depth. The five chamber depths for Configuration 4 are quarter wavelengths for the following frequencies: 1028, 1028, 1874, 2717, and 2877 Hz.

Figure 11 shows the (a) resistance, (b) reactance, and (c) absorption coefficient spectra for Configuration 4. Note that the resistance spectrum has three peaks, two of which occur near the first two antiresonances and the third of which occurs at a frequency which has a nearly asymptotic behavior in reactance. Inclusion of the facesheet increases the resistance by at least 0.5 between 800 and 2735 Hz.

Like Configuration 3, the reactance spectrum for this configuration is smoother than for the liners without facesheets. Instead of five distinct resonant frequencies, there are now three at 882, 1427, and 1835 Hz. The first resonance is followed by an antiresonance at approximately 1100 Hz.

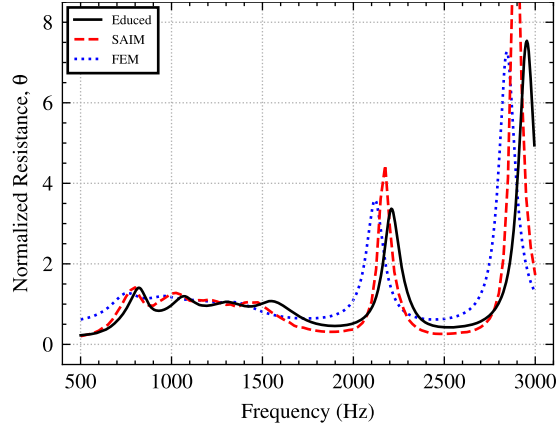
This liner provides excellent absorption. For most of the frequency range considered, the absorption coefficient is above 0.8 and is near unity at 920, 1430, 1850, and 2622 Hz. From approximately 950 to 1300 Hz and from approximately 2150 to 2500 Hz, the absorption drops below 0.8. Given these results, it seems quite reasonable to assume the sound absorption in the 1100 and 2300 Hz regions could be increased by slight modifications to the target frequencies used with the optimizer.

For this liner configuration, there are no longer five distinct resonant frequencies; inclusion of the facesheet has impacted the resistance and reactance spectra. The predicted impedance spectra show similar trends to the educed data. The SIM impedance predictions seem to compare more favorably with the educed impedances than the finite element model predictions. Recall that modifications were made to the FSM in the FEM based on Ref. [22]. These results would suggest that there could be better modifications made to some of the facesheet modeling parameters (e.g., the semi-empirical constant and the discharge coefficient) in the FEM.[¶] The educed absorption coefficients are all above 0.4 from 700 to 2800 Hz and are greater than 0.6 for most of the presented frequency range.

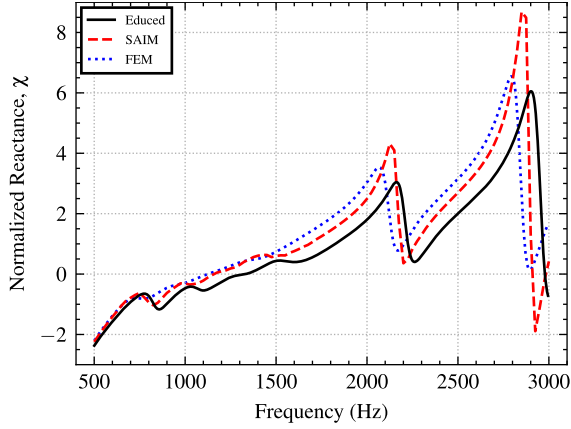
The errors in predicted impedance using both the FEM and SIM result in absorption coefficient spectra that do not completely capture the educed results. The magnitude and frequency characteristics of the measured absorption coefficient are not well captured by the SIM and the FEM predictions are even worse. However, liners designed using the optimization approach provide satisfactory results (as shown by the absorption spectra) in spite of the inadequacies of the prediction models.

Figure 12 shows a comparison of the absorption coefficient spectra between the various configurations considered for this work. Configurations 1, 2, 3, and 4 are shown in dashed-dotted green, dotted blue, dashed red, and solid black, respectively. Absorption coefficients at the target frequencies (900, 1250, 1600, 1950, and 2500 Hz) are marked with circles to highlight the absorption at these frequencies. The optimized Configuration 4 provides the best sound absorption in a broader range of frequencies compared to the other configurations. However, while the absorption at 1250 and 1600 Hz are certainly good for Configuration 4 ($\alpha > 0.8$), the absorption of the quarter-wavelength Configuration 3 liner is better at these frequencies. The absorption of Configuration 3 at higher frequencies (1950 and 2500 Hz) is poor and thus, unsuitable for an application that requires absorption in this frequency range. Comparing Configurations 1 and 2 to Configurations 3 and 4 shows the benefit of adding a facesheet on the liner's ability to absorb sound.

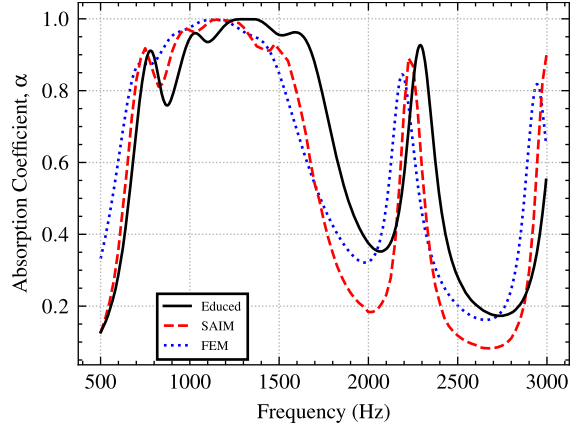
[¶]A side study revealed that the predictions from the FEM presented here matched experimental data better than when using the suggested discharge coefficient and semiempirical constant from Motsinger and Kraft [19], which suggests that some modifications to these parameters are necessary.



(a) Resistance spectrum.

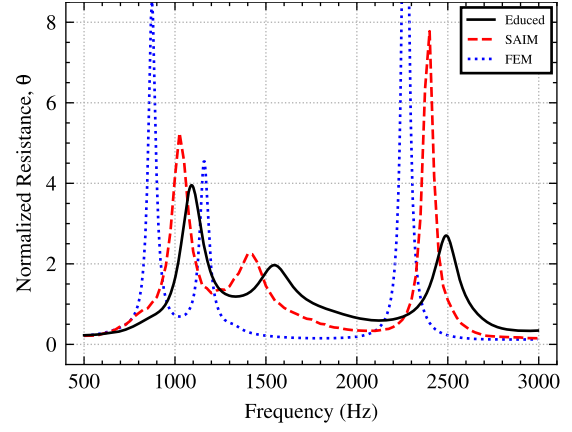


(b) Reactance spectrum.

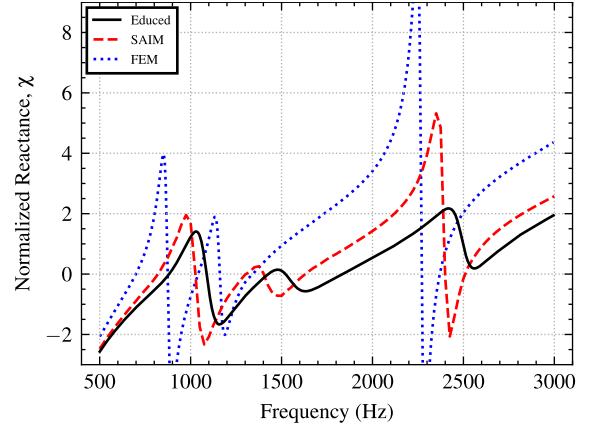


(c) Absorption coefficient spectrum.

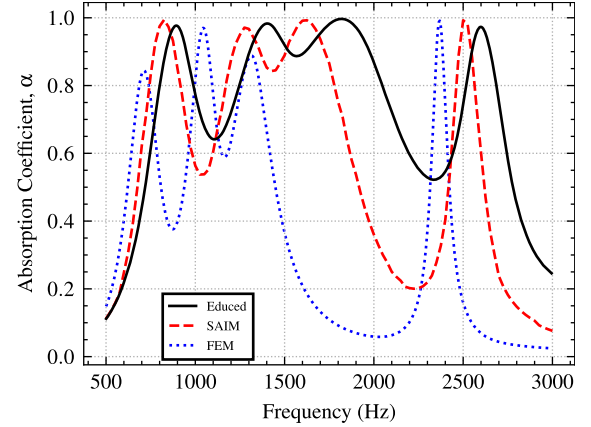
Fig. 10 Configuration 3 spectra are shown for (a) resistance, (b) reactance, and (c) absorption coefficient.



(a) Resistance spectrum.



(b) Reactance spectrum.



(c) Absorption coefficient spectrum.

Fig. 11 Configuration 4 spectra are shown for (a) resistance, (b) reactance, and (c) absorption coefficient.

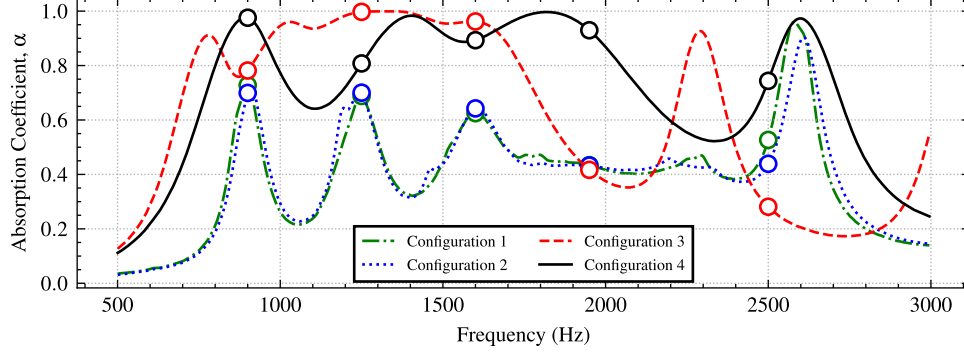


Fig. 12 Absorption coefficient spectra from Configurations 1, 2, 3, and 4. Circular markers identify the absorption at the following target frequencies: 900, 1250, 1600, 1950, and 2500 Hz.

D. Liner Performance Quantification

The L2 norm of the difference between the optimal absorption coefficient and the experimentally measured values is used to compare the unoptimized and optimized liners. In this work, the L2 norm of the difference, S , is defined as

$$S = \sum_{i=1}^{N_f} [\alpha_{opt}(f_i) - \alpha(f_i)]^2, \quad (7)$$

in which $\alpha_{opt}(f) = 1.0$ is the optimal absorption coefficient for sound attenuation and $\alpha(f)$ is the absorption coefficient calculated at a frequency f . Equation 7 is used to calculate the L2 norm at three sets of frequencies: (i) a set of five tones, (ii) from 500 to 3000 Hz, and (iii) from 700 to 2700 Hz.

When Configuration 3 was tested, the absorption coefficients at the five frequencies (900, 1250, 1600, 1950, and 2500 Hz) were 0.78, 1.00, 0.96, 0.42, and 0.28, corresponding to an L2 tone norm of 0.91. In comparison, Configuration 4 had the following absorption coefficients at the same frequencies: 0.98, 0.81, 0.89, 0.93, and 0.74, corresponding to an L2 tone norm of 0.12. Therefore, at the five frequencies, the optimized liner had an absorption coefficient that was closer to 1.0 and had a better tonal absorption than that of the quarter-wavelength liner.

While the L2 tone norm favored Configuration 4, it is of interest to investigate the L2 broadband norms for both Configurations 3 and 4. Calculating the L2 broadband norm from 500 to 3000 Hz resulted in 113.7 for Configuration 3 and 68.1 for Configuration 4. Reducing the frequency range (700 to 2700 Hz) resulted in L2 broadband norm of 196.5 for Configuration 3 and 23.7 for Configuration 4. For a broad range of frequencies, the optimized liner had an absorption coefficient closer to the optimal absorption coefficient than that of the quarter-wavelength liner, and thus, had better broadband absorption.

V. Concluding Remarks

The purpose of this work was to assess the use of an optimizer to design an acoustic liner that would target a range of tonal and broadband frequencies appropriate for a ducted propotor application. The optimizer used the SIM to design an acoustic liner. The FEM was used separately from the optimizer to provide higher fidelity predictions for comparison against the SIM.

Four acoustic liners were designed in this work: two without and two with facesheets included. All liners had an array of five chambers that were repeated five times for a total of 25 chambers. Acoustic impedances for all samples were deduced from the swept-sine analysis data acquired in the NIT.

Experimental results showed that the layout of the varied chambers affected the acoustic resistance, as predicted by the finite element model. This layout effect was not predicted by a simpler quarter-wavelength model nor by the semianalytical impedance model. For this work, this layout effect was deemed insignificant.

Although the prediction models failed to capture all of the fine details of the measured impedance for the configurations considered in this study, they were sufficiently accurate to enable design of excellent broadband absorbers. The absorption of Configuration 3, the acoustic liner designed using simple quarter-wavelength theory, is acceptable. However, Configuration 4, the optimized design, provided the best broadband absorption for the configurations considered.

There are certain limitations to the work presented. The first is that large chamber depths are required to target harmonics of the BPFs that are expected to be produced by UAM aircraft. As a result, creative packaging such as bending the chambers will be required to deploy this concept within the physical constraints of a prop rotor duct. Additional testing is required in the NASA Langley Grazing Flow Impedance Tube to assess the optimization of a variable depth liner for a prop rotor duct with grazing flow. Lastly, an investigation is required to tune the FSM parameters in the FEM to improve its prediction capabilities, which could be beneficial if the optimizer were to be coupled with the FEM instead of the SIM.

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References

- [1] Johnson, W., Silva, C., and Solis, E., "Concept Vehicles for VTOL Air Taxi Operations," *AHS International Technical Meeting on Aeromechanics Design for Transformative Vertical Flight*, American Helicopter Society, San Francisco, CA, Jan. 2018.
- [2] Silva, C., Johnson, W. R., Patterson, M. D., and Antcliff, K. R., "VTOL Urban Air Mobility Concept Vehicles for Technology Development," *2018 Aviation Technology, Integration, and Operations Conference*, AIAA AVIATION Forum, AIAA, Atlanta, GA, June 2018.
doi:10.2514/6.2018-3847, AIAA Paper 2018-3847.
- [3] Johnson, W., "A Quiet Helicopter for Air Taxi Operations," *VFS Aeromechanics for Advanced Vertical January 2020*, Vertical Flight Society, San Jose, CA, Jan. 2020.
- [4] Whiteside, S. K. S., Pollard, B. P., Antcliff, K. R., Zawodny, N. S., Fei, X., Silva, C., and Medina, G. L., "Design of a Tiltwing Concept Vehicle for Urban Air Mobility," NASA TM-20210017971, Langley Research Center, Hampton, VA, June 2021.
- [5] Whiteside, S. K. S., and Pollard, B. P., "Conceptual Design of a Tiltduct Reference Vehicle for Urban Air Mobility," *VFS Aeromechanics for Advanced Vertical Flight Technical Meeting*, Vertical Flight Society, San Jose, CA, Jan. 2022.
- [6] Wirt, L. S., "Sound-absorptive materials to meet special requirements," *The Journal of the Acoustical Society of America*, Vol. 57, No. 1, 1975, pp. 126–143.
doi:10.1121/1.380423.
- [7] Jones, M. G., Howerton, B., and Ayle, E., "Evaluation of Parallel-Element, Variable-Impedance, Broadband Acoustic Liner Concepts," *18th AIAA/CEAS Aeroacoustics Conference (33rd AIAA Aeroacoustics Conference)*, AIAA, Colorado Springs, CO, June 2012.
doi:10.2514/6.2012-2194, AIAA Paper 2012-2194.
- [8] Hua, X., Herrin, D. W., and Jackson, P., "Varying Backing Cavity Depths to Achieve Broadband Absorption using Micro-Perforated Panels," *NOISE-CON 2013*, Institute of Noise Control Engineering, Denver, CO, Aug. 2013.
- [9] Jones, M. G., Watson, W. R., Nark, D. M., and Howerton, B. M., "Evaluation of Variable-Depth Liner Configurations for Increased Broadband Noise Reduction," *21st AIAA/CEAS Aeroacoustics Conference*, AIAA, Dallas, TX, June 2015.
doi:10.2514/6.2015-2697, AIAA Paper 2015-2697.
- [10] Jones, M. G., Watson, W. R., Nark, D. M., Howerton, B. M., and Brown, M. C., "A Review of Acoustic Liner Experimental Characterization at NASA Langley," NASA TP-2020-220583, Langley Research Center, Hampton, VA, Apr. 2020.
- [11] Schiller, N. H., and Jones, M. G., "Smeared Impedance Model for Variable Depth Liners," *2018 AIAA/CEAS Aeroacoustics Conference*, AIAA, Atlanta, GA, June 2018.
doi:10.2514/6.2018-3774, 2018-3774.
- [12] Palani, S., Murray, P., McAlpine, A., Sasaki, D., and Richter, C., "Slanted septum and multiple folded cavity liners for broadband sound absorption," *International Journal of Aeroacoustics*, Vol. 20, No. 5-7, 2021, pp. 633–661.
doi:10.1177/1475472X211023835.

- [13] Palani, S., Murray, P., McAlpine, A., and Richter, C., "Optimisation of slanted septum core and multiple folded cavity acoustic liners for aero-engines," *AIAA AVIATION 2021 FORUM*, AIAA, Virtual Event, Aug. 2021.
doi:10.2514/6.2021-2172, AIAA Paper 2021-2172.
- [14] Storn, R., and Price, K., "Differential Evolution – A Simple and Efficient Heuristic for global Optimization over Continuous Spaces," *Journal of Global Optimization*, Vol. 11, No. 4, 1997, pp. 341–359.
doi:10.1023/A:1008202821328.
- [15] Nark, D. M., and Jones, M. G., "An acoustic liner design methodology based on a statistical source model," *International Journal of Aeroacoustics*, Vol. 20, No. 5-7, 2021, pp. 441–457.
doi:10.1177/1475472X211023874.
- [16] Zwikker, C., and Kosten, C. W., *Sound Absorbing Materials*, Elsevier Pub. Co, New York, 1949.
- [17] Tijdeman, H., "On the propagation of sound waves in cylindrical tubes," *Journal of Sound and Vibration*, Vol. 39, No. 1, 1975, pp. 1–33.
doi:10.1016/S0022-460X(75)80206-9.
- [18] Jones, M. G., Simon, F., and Roncen, R., "Broadband and Low-Frequency Acoustic Liner Investigations at NASA and ONERA," *AIAA Journal*, Vol. 60, No. 4, 2022, pp. 1–20.
doi:10.2514/1.J060862.
- [19] Motsinger, R. E., and Kraft, R. E., "Chapter 14: Design and Performance of Duct Acoustic Treatment: Chapter 14," *Aeroacoustics of Flight Vehicles: Theory and Practice: NASA-RP-1258-VOL-1*, edited by H. H. Hubbard, Aeroacoustics of Flight Vehicles: Theory and Practice, NASA Langley Research Center, Hampton, VA, Aug. 1991, pp. 165–206.
- [20] Parrot, T. L., and Jones, M. G., "Chapter 6: Uncertainty in Acoustic Liner Impedance Measurement and Prediction," *Assessment of NASA's Aircraft Noise Prediction Capability: NASA-TP-2012-215653*, edited by M. D. Dahl, Glenn Research Center, Cleveland, OH, July 2012, pp. 157–204.
- [21] COMSOL Multiphysics®, Software Package, Ver. 6.0, COMSOL AB, Stockholm, Sweden, 2021.
- [22] Schiller, N. H., Jones, M. G., and Bertolucci, B., "Experimental Evaluation of Acoustic Engine Liner Models Developed with COMSOL Multiphysics," *23rd AIAA/CEAS Aeroacoustics Conference*, AIAA AVIATION Forum, AIAA, Denver, CO, June 2017.
doi:10.2514/6.2017-4186.
- [23] Chung, J. Y., and Blaser, D. A., "Transfer function method of measuring in-duct acoustic properties. I. Theory," *The Journal of the Acoustical Society of America*, Vol. 68, No. 3, 1980, pp. 907–913.
doi:10.1121/1.384778.
- [24] Jones, M. G., "An improved model for parallel-element liner impedance prediction," *3rd AIAA/CEAS Aeroacoustics Conference*, AIAA, Atlanta, GA, May 1997, pp. 519–526.
doi:10.2514/6.1997-1649, AIAA Paper 97-1649-CP.
- [25] Watson, W. R., and Jones, M. G., "A Comparative Study of Four Impedance Eduction Methodologies Using Several Test Liners," *19th AIAA/CEAS Aeroacoustics Conference*, AIAA, Berlin, Germany, May 2013.
doi:10.2514/6.2013-2274, AIAA Paper 2013-2274.
- [26] Howerton, B. M., Vold, H., and Jones, M. G., "Application of Swept Sine Excitation for Acoustic Impedance Eduction," *25th AIAA/CEAS Aeroacoustics Conference*, AIAA, Delft, The Netherlands, May 2019.
doi:10.2514/6.2019-2487, AIAA Paper 2019-2487.
- [27] Brown, M. C., and Jones, M. G., "Evaluation of Variable Facesheet Liner Configurations for Broadband Noise Reduction," *AIAA AVIATION 2020 FORUM*, AIAA, Virtual Event, June 2020.
doi:10.2514/6.2020-2616, AIAA Paper 2020-2616.
- [28] Jones, M. G., and Howerton, B. M., "Benchmark Data for Evaluation of NASA Impedance Eduction Methods," *AIAA AVIATION 2021 FORUM*, AIAA, Virtual Event, Aug. 2021.
doi:10.2514/6.2021-2143, AIAA Paper 2021-2143.